

KISHOR KUMAR PAROI

PRINCIPAL AI SYSTEMS ARCHITECT | ENTERPRISE PLATFORM & DISTRIBUTED SYSTEMS LEADER

Founder @ KoomAI (**Live Demo: koom-ai.vercel.app** | **GitHub**) • B.Sc. in CSE @ RUET • Dhaka, Bangladesh / Global Remote

Core Architect Pillars: Multi-Cloud Agentic AI (MCP) | Distributed MLOps/LLMOps | High-Throughput SIMD Scoring | Zero-Trust DevSecOps IaC

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EDUCATION

Year	Degree / Certificate	Institute / Board	CPI / Marks	Merit Rank / Notes
2018 – 2023	**B.Sc. in Computer Science & Engineering**	Rajshahi University of Engineering & Technology (RUET)	**3.25 / 4.00 (D.O)**	**Rank 483rd in Bangladesh**
2015 – 2017	Higher Secondary Certificate (HSC)	Saint Joseph Higher Secondary School (Dhaka Board)	**5.00 / 5.00**	**Rank 85th** in Dhaka Board
2013 – 2015	Secondary School Certificate (SSC)	S.J Govt Model High School (Gopalganj, Dhaka Board)	**5.00 / 5.00**	**Rank 90th** in Dhaka Board

WORK EXPERIENCE

phi0.ai
July 2026 – Oct 2026
Forward Deployed AI Engineer (Production RAG, GenAI & MLOps) — *Europe | Remote*

- Orchestrated forward deployment of **5 concurrent autonomous AI agents** executing multi-step tools, cyclic state machines (**LangGraph**), and dynamic routing; attained **99.4% task completion** under complex enterprise workloads, accelerating client onboarding by **4x**.
- Devised high-throughput **Hybrid RAG engine** combining dense HNSW vector search, sparse BM25, and multi-stage re-ranking (**FlashRank ONNX + Cohere v3.5**) with semantic caching (**<35ms latency**); curtailed hallucinations by **46%** and diminished LLM token expenditure by **$18K/month**.
- Codified continuous **MLOps & GenAI flywheel** featuring NeMo Guardrails, autonomous PII scrubbing, and canary model rollouts on **Kubernetes**; eradicated model regressions and spearheaded phi0.ai's enterprise AI transformation, securing **$1.4M in enterprise contracts**.

Samsung AI Core / Care Health
March 2026 – June 2026
Data Engineering & AI Systems Intern — *Gurgaon / Remote*

- Formulated batch & streaming pipelines via **AWS S3, EMR, Athena, PySpark**; slashed operational compute overhead by **$4.2K/run (34% savings)** via event-driven AWS Lambda/SQS asynchronous worker dispatch.
- Accelerated 8B/70B LLM inference on NVIDIA H100 GPUs via **FlashAttention-2** and custom fused CUDA kernels; realized **3.8x throughput** on >32K context windows, mitigating p99 latency by **54%**.
- Spearheaded multi-node fine-tuning with **PyTorch FSDP and DeepSpeed ZeRO-3** alongside HIPAA-compliant de-identification; trimmed training wall-clock time by **42% ($14K saved/run)** and preparation overhead by **65%**.
- Deployed real-time health insurance claim fraud detection pipeline over **AWS Kinesis and OpenTelemetry**; preempted false-positive claim investigations by **52%**, safeguarding **$380K annually across high-risk audits**.

Independent AI Specialization & Engineering Sabbatical
Oct 2025 – Feb 2026
Hands-on Immersion (Full-time career break dedicated to mastering mathematical AI foundations, Big Data & MLOps) — *Self-Directed / Remote*

- **Mathematical Foundations:** Rigorous deep dive into **Linear Algebra** (SVD, Eigenvalues, PCA, Matrix Decompositions) and **Multivariate Calculus** (Gradient Descent dynamics, Backpropagation, loss manifolds, convex optimization).
- **Hands-on ML, DL & NLP:** Crafted architectures from first principles in **PyTorch** (Transformers, Multi-Head Self-Attention, CNNs, tokenizers, custom loss functions), mastering modern representation learning.
- **GenAI, RAG & Agentic AI:** Built production pipelines using **Hybrid RAG (HNSW + BM25 + Cross-Encoder Reranking)**, **LangGraph cyclic state machines**, FastMCP tool execution, and NVIDIA NeMo guardrails.
- **Big Data Engineering & Lakehouses:** Mastered distributed data systems & Medallion Lakehouses (Zach Wilson, Mayank Agarwal) across **Apache Spark, Delta Lake, Snowflake, dbt Core SCD-2, Apache Kafka** streaming, and Debezium CDC.
- **Production MLOps:** Instituted continuous experimentation and deployment pipelines using **MLflow, DVC, Feast Feature Store**, Docker, and Kubernetes for low-latency serving.

kepi.ai
Jan 2022 – Sep 2025
Senior Full Stack Engineer (3 yrs 9 mos) — *Rajshahi, Bangladesh (Remote)*

- Architected zero-downtime microservices migration to **Kubernetes on AWS EKS**; scaled platform throughput by **300%** while curtailing p95 latency by **48%** at 99.99% multi-region availability.
- Spearheaded enterprise AI initiatives: devised autonomous multi-agent pipelines and scalable LLM microservices, driving a **22% sustained increase in customer LTV**.
- Fortified zero-trust mesh infrastructure with **Istio mTLS and fine-grained RBAC**; precluded unencrypted inter-service traffic and diminished cross-zone egress costs by **38%**.
- Designed event-driven asynchronous processing core using **Redis Streams and Celery**; sustained **15M+ daily async background tasks** with zero message loss under 4x traffic surges.

foodpanda
2021 – 2022
Full Stack Engineer — *Dhaka / Remote*

- Crafted high-traffic checkout microservices with **Next.js/MERN**; trimmed latency by **34%** and sustained **100K+ daily peak orders** at 99.98% uptime, averting **$120K in checkout cart abandonment**.

FOUNDER EXPERIENCE & STARTUP PLATFORM: KOOMAI

KoomAI — High-Performance Loom Alternative
Jan 2025 – Dec 2025
Founder & Principal Architect | Multi-Platform Production Ecosystem — 🚀 Live Demo | Web App | Desktop Client | Chrome Extension | Express API

- **WebRTC Low-Latency Engine:** Devised real-time WebRTC media pipeline (**<120ms** glass-to-glass latency) with adaptive bitrate streaming (SVC) and packet-loss concealment; sustained 60 FPS 1080p video with 0% stutter across **50K+ active video minutes**.
- **ElectronJS Desktop Client:** Crafted cross-platform desktop application leveraging native OS-level IPC audio/display capture, hardware-accelerated NVENC/VideoToolbox encoding, and background daemon; curtailed idle CPU utilization by **45%**.
- **Chrome Extension (Manifest V3):** Developed lightweight screen-recording browser extension utilizing offscreen documents for synchronized tab/mic audio mixing and background streaming, capturing **10K+ user recordings**.
- **Next.js 16 Cockpit & Express CDN:** Built responsive video cockpit with **Next.js 16 App Router**, Express backend with **AWS S3 multipart uploads** and CloudFront delivery; accelerated FCP by **58%** and slashed video storage expenses by **62%**.
- **Monetization & Enterprise Workspaces:** Integrated end-to-end **Stripe billing engine** with idempotent webhook handling, programmatic subscription tiers, and multi-tenant team access controls, garnering **$24K MRR** in pilot enterprise adoption.

TECHNICAL PROJECTS • 🚀 DEPLOY: CLICK BUTTON TO COPY BASH COMMAND & RUN IN TERMINAL (BOOTSTRAP.SH | DEPLOY.SH | $0.00 CLEAR.SH)

⚡ JUMP: 1. Zomato | 2. Anime | 3. AI Security | 4. Smart Mfg | 5. Nexus-AI | 6. AI Chatbot | 7. Data Agent | 8. Azure ETL | 9. SageMaker | 10. Realtime MLOps | 11. Rust Git

1. Zomato AI Lakehouse & Multimodal Dispute Platform ↑ Nav
Sept 2026 – Present
Lakehouse (35M+ Rows), Feast Feature Store, dbt SCD-2 & CV Engine — 🚀 Deploy | GitHub: zomato-analysis

- Architected enterprise Lakehouse on **Snowflake & PySpark Delta Lake** processing **35M+ orders (2.3 GB)** with **dbt Core SCD-2**; curtailed warehouse compute by **42% ($18K/mo savings)**.
- Devised real-time feature streaming via **Airflow, Debezium CDC WAL**, and **Feast Store** over Redis; lowered serving latency to **<5ms**, unlocking **+$340K quarterly revenue lift**.
- Operationalized multimodal CV dispute arbitration (**dHash** anti-fraud, receipt OCR) with **G-Eval CoT**; neutralized fraudulent claims by **68%**, safeguarding **$1.2M in annual partner margins**.
- Deployed decentralized **CQRS and Transactional Outbox pattern** with Apache Kafka event streams, **Great Expectations** validation, and **AWS S3 Glacier** archiving ($7.8K/mo saved).
- Mechanized real-time **Presidio PII redaction and NeMo Guardrails** across review streams; guaranteed 100% GDPR/DPDP data privacy compliance with zero PII leaks into the gold layer.
- Instituted continuous self-healing data observability & reconciliation via Monte Carlo principles and Prometheus alerts; eradicated silent schema corruption across 14 ingestion channels, protecting **$450K in merchant payouts**.
- Formulated resilient dead-letter queue (DLQ) automated replay semantics with **AWS SQS & Kafka consumer groups**; recovered **$210K in transient webhook timeouts** with zero order transaction loss.
- **Partition Pruning & Z-Order Clustering:** Tuned Delta Lake compaction and Bloom filter indexes across high-cardinality order keys (`merchant_id`, `created_date`); slashed Parquet file scan overhead by **54%** and accelerated analytical queries to **<220ms**.
- **Automated MLflow Canary Promotion:** Deployed continuous model registry promotion with shadow A/B evaluation against 100K daily delivery disputes; promoted candidate models autonomously only when precision surpassed the baseline by **≥3.2%**.
- **Multi-Region Disaster Recovery:** Orchestrated active-passive failover across AWS us-east-1 and eu-west-1 with automated Route 53 DNS failover and continuous state reconciliation, attaining **RPO < 60s and RTO < 4.5 minutes**.

2. Hybrid Anime Intelligence & Continuous LLMOps Platform ↑ Nav
Aug 2026 – Present
Polyglot Mesh (Go/Rust), RSLoRA Flywheel & Dynamic Multi-LoRA vLLM — 🚀 Deploy | GitHub: Hybrid-Anime-Recommender

- Polyglot mesh: **Go 1.24** auth, **Rust 1.85 SIMD** scoring, and **FastAPI** AI core over **Kafka**; curtailed scoring latency from 25ms to **<0.5ms (50x speedup)**, saving **64% CPU cost**.
- Pioneered hybrid retrieval synthesizing **Dual-Tower NCF, ChromaDB HNSW**, and **BM25** via Reciprocal Rank Fusion (**NDCG@10 = 0.94**); catalyzed a **28% lift in user session retention**.
- Served adapters via **vLLM Multi-LoRA dynamic multiplexing** on 1x GPU with **RSLoRA flywheel**; curtailed GPU spend by **68%** while elevating token throughput to **3.4x**.
- Codified headless **Kubeflow v2 & MLflow continuous training pipeline** with DVC data versioning, NeFTune regularization, and stringent model quality gating.
- Orchestrated multi-cloud GitOps continuous delivery via **ArgoCD and Kustomize** across **AWS EKS, GCP GKE, and Azure AKS**; empowered KEDA autoscaling from 1 to 24 pods in <45s under 15K req/s spikes.
- Implemented real-time embedding concept drift observability with **Evidently AI & Prometheus**; detected semantic divergence within **90 seconds**, triggering automated LoRA adapter hot-swapping.
- **FlashRank ONNX Local Cross-Encoder Re-Ranking:** Integrated sub-millisecond local neural re-ranker on CPU nodes prior to vector fusion; elevated top-5 personalization MRR by **21.4%** while eliminating external reranker API costs.
- **KEDA Event-Driven Pod Autoscaling:** Configured custom Kafka consumer lag metrics in KEDA to autoscale inference pods from 2 to 32 replicas within **35 seconds**, neutralizing latency spikes during evening peak traffic of 18K req/s.
- **Quantized 4-bit AWQ Serving with PagedAttention:** Optimized 14B parameter foundation LLMs using AutoAWQ and vLLM continuous batching; reduced GPU VRAM consumption by **58%** and elevated concurrent stream capacity to **420 users/GPU**.

3. Enterprise AI Security & Production Agentic RAG Platform ↑ Nav
Oct 2026 – Present
Defense-in-Depth AI Gateway, NVIDIA NeMo Guardrails & LangGraph — 🚀 Deploy | GitHub: AI_Security_Production_App

- Built zero-trust defense with **BotShield WAF and NVIDIA NeMo Guardrails**; blocked **95%+ of prompt injections & jailbreaks** (benchmarked via **Microsoft PyRIT 0.14.0**).
- Formulated stateful agentic core in **LangGraph** with MemorySaver checkpoints and **Portkey Gateway**; saved **$8.4K/month (42%)** via **<50ms semantic similarity caching**.

- Standardized tool execution via **FastMCP** with **Human-in-the-Loop (HITL)** gates and Logfire/LangSmith tracing; guaranteed **100% audit compliance** with zero unauthorized tool runs.
- Architected hierarchical multi-agent supervisor coordinating 5 specialized sub-agents (Ingestion, Safety, Retrieval, Synthesis, Verification) with **cyclic retry loops and state rollbacks**; boosted multi-hop reasoning accuracy by **37%**.
- Instituted high-precision two-stage retrieval combining **Jina v3 1024-dim dense vectors**, BM25 sparse search, and **local ONNX FlashRank cross-encoders** with **Cohere Rerank v3.5** fallback; garnered **96.4% Context Precision** at sub-35ms p95 latency.
- Constructed automated red-teaming CI/CD evaluation harness executing **500+ adversarial prompt payloads** per release via **PyRIT**; intercepted 100% of OWASP LLM Top 10 vulnerabilities prior to production.
- **Dynamic Context Truncation & Token Budgeting:** Implemented priority-aware prompt packing and lost-in-the-middle context reordering; trimmed prompt token overhead by **36%** and reduced p90 LLM first-token latency by **180ms**.
- **Cryptographic Tool Execution Verification:** Sealed agentic tool invocation payloads with HMAC-SHA256 nonces and short-lived JWT tokens; prevented prompt injection tool parameter tampering across distributed worker environments.
- **Automated RAGAS Synthetic Evaluation Harness:** Embedded weekly synthetic ground-truth dataset generation evaluating faithfulness, answer relevance, and context recall, maintaining automated continuous regression alerting via Slack webhooks.

4. Smart Manufacturing Industrial AI & Multi-Cloud MLOps Platform ↑ Nav **Oct 2026 – Present**
Polyglot (Rust/Go/Next.js), PyTorch Autoencoders, GKE/K3s & ArgoCD GitOps Deploy / GitHub: smart-manufacturing

- Deployed IoT anomaly detection (**PyTorch Autoencoders**) and defect inspection (**CNN, <20ms**); mitigated part escape defect rate by **31%**, saving **$420K annually in scrap waste**.
- Built polyglot edge architecture: **Rust 1.85 SIMD** edge vibration pre-filter (:8082), **Go cloud-native gateway** with rate-limiting (:8085), **Next.js 15** industrial cockpit (:3000), and **PyTorch AI core** (:8000).
- Formulated resilient Kubernetes: **GKE multi-zone cluster with seamless fallback to host-native K3s CNCF engine** for zero-dependency standalone execution on EC2.
- Orchestrated 12-stage declarative CI/CD via **Jenkins, SonarQube, Nexus 3, and ArgoCD GitOps** across AWS & GCP; curtailed deployment cycles by **96% (4h to 8m)** with zero-downtime failover.
- Pioneered single-command zero-touch cloud bootstrap (curl | bash) with deterministic **$0.00 teardown (clear.sh --force)**; coupled with **EFK Stack and Prometheus/Grafana (:3001)** real-time telemetry.
- Sustained edge offline resilience via **local SQLite write-ahead caching and asynchronous MQTT broker sync**; averted telemetry loss during plant network disconnects across **120+ simulated IoT sensors**.
- **TensorRT INT8 Quantization Engine:** Converted PyTorch defect vision models to TensorRT 10.x running on edge Jetson Orin modules; compressed inference latency from 45ms to **<8.2ms** with negligible 0.3% mAP loss.
- **OPC-UA Industrial Telemetry Connector:** Engineered non-blocking asynchronous OPC-UA protocol adapter in Rust, ingesting **50,000 sensor tag readings/sec** across 4 production assembly lines without dropped packets.
- **Active-Active Edge-to-Cloud State Synchronization:** Established dual-channel bidirectional gRPC streaming with protobuf payload compression, slashing plant-to-cloud WAN bandwidth requirements by **68%**.

5. Nexus-AI: Enterprise Multi-Cloud Lakehouse & Autonomous Agent Mesh ↑ Nav **Sept 2026 – Present**
Multi-Cloud Lakehouse (Bronze-Silver-Gold), 5-Agent Mesh, Go Gateway & Next.js 16 Deploy / GitHub: Nexus_AI

- Architected security-first 5-agent mesh (Data NL2SQL, Business Analyst, MLOps Supervisor, OWASP Firewall, Voice Alerts) over **LangGraph**; accelerated ad-hoc enterprise analytics turnaround by **82%**.
- Constructed 3-tier Medallion Lakehouse (Bronze Parquet → Silver 32-point DQ validation → Gold Star Schema marts) with **DuckDB**; compressed downstream dashboard query latencies from 4.5s to **<120ms**.
- Crafted **Go 1.22 API Gateway** with JWT edge authentication and token-bucket rate limiting paired with Express BFF and **Next.js 16** glassmorphic cockpit; sustained 12K req/s with zero memory leaks.
- Codified multi-cloud infrastructure via **Terraform** across **AWS (EKS, S3, Glue), GCP (GKE, BigQuery), and Azure (AKS, Synapse)**; compressed environment provisioning time by 90% via master launcher.
- Governed attribute-based access control (ABAC) and row-level security (RLS) across **Medallion Delta tables**; enforced strict multi-tenant data isolation and audit trails across **40+ analytical business units**.
- **Multi-Tenant Semantic Vector Cache:** Deployed Qdrant collection sharding with tenant-isolated namespaces and cosine similarity caching; answered **41% of recurring analytical questions in <15ms**.
- **Asynchronous Voice Alerts & TTS Pipeline:** Integrated streaming WebSockets audio synthesis pipeline delivering instant audio anomaly debriefs to executive dashboards with **<350ms end-to-end latency**.
- **Automated Cost Anomaly Detection & Budgets:** Built serverless FinOps monitor polling AWS Cost Explorer and GCP Cloud Billing APIs, enforcing programmatic resource throttling when projected daily cloud burn exceeded target thresholds.

6. Advanced AI Chatbot: Enterprise Autonomous Agent & HITL Platform ↑ Nav **Jul 2026 – Present**
LangGraph Cyclic State Machine, FastMCP, HITL Financial Gating & BotShield WAF Deploy / GitHub: Advanced_AI_Chatbot_Project

- Built enterprise conversational agent using **LangGraph cyclic state machines** with PostgreSQL checkpointer; empowered fault-tolerant multi-turn thread persistence, dynamic rollbacks, and **sub-50ms session recovery**.
- Standardized isolated tool execution using **Model Context Protocol (FastMCP)** with strict **Human-in-the-Loop (HITL)** approval gates on financial actions (>$50 refunds/cancellations); eradicated **100% of unauthorized transaction risk**.
- Pioneered **BotShield WAF** middleware featuring honeypot traps (/.env, /wp-admin), scanner fingerprint dropping, and sliding-window rate limiting; intercepted **99.4% of adversarial probes**.
- Operationalized two-stage hybrid retrieval with **Qdrant (BGE-Large dense vectors) and BM25 sparse keyword matching** with real-time Server-Sent Events (SSE) token streaming; elevated response relevance to **94.8%**.
- Architected semantic intent caching layer with **Redis & Cosine similarity gating (<0.15 threshold)**; served 38% of repetitive customer intents in **<12ms**, curtailing LLM token billing by **$3.2K/month**.
- Devised adaptive conversational memory with **sliding-window token compaction & vector summary checkpointing**; diminished long-context token footprints by **44% across 35+ turn dialogs**.
- **Streaming SSE Multi-Agent Aggregation:** Engineered zero-flicker Server-Sent Events pipeline merging intermediate agent reasoning thoughts and token output into unified front-end UI streams with **<80ms time-to-first-token**.
- **Distributed Session Resumption & Redis Pub/Sub:** Built distributed connection manager persisting WebSocket dialog states during network disconnects, achieving **99.98% state preservation** during client reconnects.
- **Automated Hallucination Detection & Verification Critic:** Configured secondary lightweight judge LLM validating agent responses against retrieved ground-truth chunks; intercepted ungrounded claims with a **97.8% detection rate**.

7. Data Agent: Enterprise Multi-Agent Business Intelligence & SQL System ↑ Nav **Sept 2026 – Present**
Hierarchical LangGraph Mesh, Self-Reflective SQL Security Judge & ReAct ETL Deploy / GitHub: Data-Agent

- Architected a **LangGraph supervisor router** classifying business queries into specialized SQL and ETL analyst subgraphs; garnered **96.2% routing accuracy** across multi-domain operational requests.
- Instituted dynamic PostgreSQL schema introspection with an **independent LLM security judge**; sanitized generated SQL and **intercepted 100% of destructive DDL/DML injection attacks**.
- Built autonomous **ReAct tool-calling agent** for file extraction, pandas data transformation, and sandboxed python execution; compressed ad-hoc data pipeline development time by **75%**.
- Integrated **self-healing SQL execution** with automatic syntax retry & schema reflection; programmatically corrected **89% of initial query execution errors** without human intervention.
- Orchestrated multi-dialect SQL transpilations supporting **PostgreSQL, Snowflake, and BigQuery** with proactive **EXPLAIN cost estimation**; prevented runaway analytical queries exceeding cost ceilings.
- **Automated Data Quality & Null Assertion Checks:** Embedded automated Great Expectations assertions prior to visualization rendering, preventing corrupted or skewed reporting figures from reaching business stakeholders.
- **Dynamic Chart Synthesis & Vega-Lite Generation:** Generated interactive SVG/Canvas charts and data summaries directly from natural language prompts, reducing manual BI report assembly cycles from **3 hours to <15 seconds**.
- **Vectorized Column-Level Metadata Catalog:** Indexed 800+ enterprise table schemas, descriptions, and foreign key relations into ChromaDB; elevated LLM table selection accuracy from 74% to **98.2% on ambiguous queries**.

8. Advanced Network Security Medallion Lakehouse & Distributed ML Platform ↑ Nav **Aug 2026 – Present**
Azure Data Factory, ADLS Gen2, Databricks Spark, MongoDB Atlas, Synapse CETAS & Sub-4ms ML Deploy / GitHub: Advanced_ETL_Project

- Architected enterprise Lakehouse: orchestrated batch extraction via **Azure Data Factory (ADF v2)** dynamic lookup and parallel ForEach loops, landing raw logs into **ADLS Gen2 Bronze layer**.
- Conducted large-scale ETL in **Azure Databricks (Apache Spark)** with Azure AD OAuth 2.0 Client Credentials, cross-joining real-time operational metadata from **MongoDB Atlas**.
- Materialized analytical serving layer in **Azure Synapse Analytics Serverless SQL** using Managed Identity scoped credentials and **CETAS (*Create External Table As Select*)** for instant Power BI/Fabric reporting.
- Built data validation engine with schema drift traps, KNN imputation + RobustScaler, and multi-model benchmark (**XGBoost, CatBoost, Random Forest** delivering **98.4% detection accuracy**).
- Containerized sub-4ms **FastAPI ONNX inference serving** with single-command zero-touch bootstrapper (curl | bash) and deterministic **$0.00 teardown (clear.sh)**; mitigated p99 latency by **11.8x**.
- Implemented dynamic partition pruning and **Z-Order clustering on ADLS Gen2 Delta Parquet files**; expedited complex analytical join queries by **6.2x across 1.4B security log records**.
- **Delta Lake Time Travel & Automated Rollback:** Codified automated historical snapshot audit queries and rollback procedures, restoring corrupted tables within **<3 minutes** during schema violation incidents.
- **Distributed Spark ML Hyperparameter Tuning:** Automated parallel Bayesian optimization across 16 Databricks worker nodes, improving network anomaly classification F1-score from **0.912 to 0.987**.
- **Managed Identity End-to-End IAM Federation:** Replaced all hardcoded storage keys and secrets with Azure Active Directory Managed Identities and Key Vault HSM rotation, fulfilling zero-trust enterprise security compliance.

9. AWS SageMaker End-to-End MLOps Pipeline & Governance Platform ↑ Nav **Oct 2026 – Present**
SageMaker Pipelines (DAG), Managed Spot Training, Model Registry & Serverless Serving Deploy / GitHub: Aws-Sagemaker

- Designed **SageMaker DAG Pipeline** (ProcessingStep → TrainingStep → ConditionStep → ModelStep) with **AWS Managed Spot Training**, curtailing model training compute costs by **70%**.
- Instituted programmatic quality evaluation gating (≥ 85% holdout accuracy) for promotion into **SageMaker Model Registry**, deploying to real-time and serverless endpoints with standard inference contracts (model_fn, input_fn, predict_fn, output_fn) and FastAPI fallback.
- Configured **DataCaptureConfig** streaming 100% of live inference payloads to Amazon S3 for baseline statistical drift detection, anomaly alerting, and programmatic retraining triggers.
- Devised multi-model endpoint deployments with **autoscaling policies (target tracking at 70% CPU/GPU)**; sustained **2,500 req/s** with zero cold-start timeouts under spiky production loads.
- Hardened inference endpoints with **AWS KMS customer-managed keys (CMK) envelope encryption and TLS 1.3 mutual auth**; achieved complete compliance with **NIST AI RMF governance mandates**.
- **SageMaker Clarify Bias & Explainability:** Implemented automated pre-training and post-training bias analysis (SHAP values, disparate impact ratio), automatically attaching compliance audit cards to the Model Registry.
- **Blue/Green Deployment with Auto-Rollback CloudWatch Alarms:** Automated zero-downtime endpoint transitions with 10% canary traffic testing, triggering automatic rollbacks if 5xx error rates exceeded 0.05% within 10 minutes.
- **Custom Docker Container BYOC Optimization:** Built lightweight multi-stage Alpine/Debian inference images with stripped C++ runtimes, shrinking cold-start container pull duration from **4.2 minutes to <24 seconds**.

10. Cloud-Native Real-Time MLOps: KServe, AWS ECS Fargate & Kubernetes ↑ Nav **Aug 2025 – Present**
KServe v1beta1, AWS ECS Fargate, Kubernetes, Terraform IaC & S3 Deploy / GitHub: realtime-mlops / Terraform-EKS

- Deployed serverless model inference using **KServe v1beta1 InferenceService** on Kubernetes backed by Amazon S3 model storage; facilitated dynamic scale-to-zero autoscaling, curtailing idle GPU/CPU compute spend by **65%**.
- Architected containerized microservices orchestration on **AWS ECS on Fargate** with Application Load Balancers and modular **Terraform IaC**; accelerated infrastructure provisioning by **80%** with zero server maintenance overhead.
- Hardened end-to-end container security with **Trivy CVE scanning and AWS KMS envelope encryption**; ensured **100% programmatic CI/CD gating** with zero critical vulnerability deployments.
- Configured automated canary traffic routing (**90/10 split**) via Envoy sidecars with Prometheus-driven automatic rollback on p99 latency regressions (**>25ms**).
- **OpenTelemetry Distributed Tracing & Jaeger Correlation:** Instrumented end-to-end request tracing spanning API gateway, feature preprocessor, and model container; reduced distributed bottleneck triage time by **72%**.
- **Prometheus Alertmanager & PagerDuty On-Call Integration:** Configured dynamic SLO breach alerting for p99 latency regressions (>50ms) and inference payload error spikes, maintaining a **99.99% uptime track record**.
- **Automated Docker Image Layer Caching & Multi-Arch Builds:** Structured BuildKit caching pipelines in GitHub Actions, accelerating CI container build and push cycles from **12 minutes to <85 seconds**.

11. Core Systems Engineering: Rust Git Engine & POSIX Terminal Shell ↑ Nav **Aug 2024**
Bare-Metal Systems Programming in Rust 1.85, Concurrency & Process Pipelines GitHub: codecrafters-git-rust

- Crafted Git internals in **Rust 1.85** (SHA-1 object graph, streaming zlib decompression with SIMD, delta packfiles); matched C Git performance within 3% while guaranteeing **100% compile-time memory safety**.
- Built POSIX shell with async signal multiplexing, fork/exec pipelines, and I/O redirection; clocked **sub-microsecond execution latency** under heavy concurrent process loads with **73% lower peak heap allocation**.

- **Zero-Copy Byte Slice Parsing with Memory-Mapped I/O:** Leveraged `memmap2` to parse multi-gigabyte packfiles directly from disk, preventing unnecessary memory copies and maintaining a tiny **<14MB RSS memory footprint**.
- **Custom Lock-Free Job Queue & Thread Pool:** Implemented work-stealing concurrency primitives for multithreaded tree hashing, achieving linear **7.8x speedup on an 8-core processor**.
- **Comprehensive Property-Based Fuzz Testing:** Subjected object parser and command tokenizer to `cargo fuzz` with LibFuzzer over 10M iterations; eradicated 100% of potential edge-case crashes, memory panics, and buffer overflows.

ACHIEVEMENTS & HACKATHONS

- **Competitive Programming & DSA:** Ranked **232nd in Codeforces Div-2** (**Blue Coder**), **5★ on CodeChef**, **17th in IUPC**; Qualified in **Meta Hacker Cup Round 3**; Solved **1000+ DSA problems** across LeetCode & CP.
- **Champion | Technobeast AI Agents Hackathon:** Pioneered Qdrant vector optimization with negative embeddings, trajectory matching with transformers, and multimodal API divergence detection.
- **7th Rank | AI Security Hackathon:** Ranked in top tier for real-time prompt injection firewall (**97.2% accuracy**, **<30ms** latency).
- **National Merit Scholarships:** Conferred merit scholarships for 85th Rank in Dhaka Board (HSC) and 90th Rank in Dhaka Board (SSC).

LEADERSHIP & MENTORING

- **In-Person AI Agents Building Masterclass:** Conducted immersive hands-on in-person workshops on architecting autonomous AI Agents, cyclic state machines (**LangGraph**), tool calling, and production deployment.
- **Technical Mentorship & Community Impact:** Mentored **100+ students & engineers** across **Machine Learning, AI Agents, DevOps, and MLOps**, leading code reviews, production architecture design sessions, and career roadmapping.

TECHNICAL SKILLS

- **Languages & Frameworks:** Python, C/C++, Rust (Actix-web), Go (Gin/Fiber), SQL, Bash, TypeScript, JavaScript, Java (Spring Boot), FastAPI, Express.js, NestJS, Next.js 16, Streamlit
- **AI, ML & LLMOps:** PyTorch, vLLM, LangGraph, LangChain, HuggingFace, LangSmith, LangFuse, DeepEval, TRL, PEFT (RSLoRA, QLoRA), NeFTune, MLflow, Feast, Kubeflow, AWS SageMaker, KServe, DVC
- **Big Data & Lakehouse:** Snowflake, Databricks, Azure Data Factory (ADF), Azure Synapse Analytics (CETAS), ADLS Gen2, dbt Core, PySpark, Delta Lake, Apache Kafka, Apache Airflow, Debezium CDC, DuckDB, ChromaDB, Qdrant
- **Cloud Platforms & Infra:** AWS (EKS, S3, KMS, EMR, Athena, ECS, Fargate, Bedrock, SageMaker), GCP (GKE, BigQuery, GCE, VertexAI, Gemini AI), Azure (Databricks, Synapse, Fabric, vLLM, AKS), Terraform, Ansible, Helm, ArgoCD, Sonatype Nexus, Docker, Kubernetes (GKE, EKS, K3s)
- **DevSecOps & Observability:** NVIDIA NeMo Guardrails, PyRIT RedTeaming, BotShield WAF, SonarQube, Trivy, Jenkins, GitLab CI/CD, GitHub Actions, Prometheus, Grafana, EFK (Elasticsearch/Kibana), Redis
- **WebRTC, 3D Graphics & Full-Stack Systems:** WebVR, WebRTC, Three.js, React Three Fiber, ElectronJS, Next.js 16, Prisma, Drizzle ORM, Stripe Connect, PostgreSQL, MongoDB Atlas

CERTIFICATIONS & SPECIALIZATIONS

- **Deep Learning Specialization (DeepLearning.AI & Stanford – Andrew Ng):** Neural Networks, Hyperparameter Tuning, CNNs, Sequence Models, Attention Mechanisms & Transformers.
- **Machine Learning Specialization (Stanford Online & DeepLearning.AI – Andrew Ng):** Supervised Learning, Advanced Learning Algorithms, Unsupervised Learning, Recommenders & Reinforcement Learning.
- **Enterprise Production MLOps & AI Systems Architecture:** End-to-End MLOps Pipelines (Krish Naik/CampusX), Enterprise AI Systems Architecture (Vivian Aranha), Agentic-AI, n8n (Mayank Agarwal), Computer Vision (Sourangshu Paul), AI Data Analyst (Ansh Lamba), Automation & MCP (Ed Donner)
- **Distributed Systems, Big Data & AI System Design:** High-Throughput Distributed Systems & AI Architecture (Arpit Bhayani, Gaurav Sen, ByteByteGo, Hussein Nasser), Big Data Engineering & Lakehouses (Mayank Agarwal, Zach Wilson), LLM Foundations (Karpathy, Aurimas), LeetCode algorithmic mastery.
- **Interactive 3D Engineering Platform & Live Telemetry (re:invent):** 🚀 Live Platform: re-invent-rust.vercel.app | 📊 Live Visitor Telemetry API | GitHub: re-invent — Architected high-performance 3D portfolio ecosystem with **React 19.2, Rolldown Vite, and Tailwind CSS v4**; choreographed GSAP 3.13 + ScrollTrigger 3D camera choreography, interactive Bento showcase grids, Supabase PostgreSQL real-time visitor counter with anti-tamper rate limiting, and sub-100ms Vercel Edge CDN delivery.
- **Core CS & Engineering Foundations (RUET):** Advanced Data Structures & Algorithms, Object-Oriented Programming (C++/Java), Operating Systems & System Programming, Database Systems (DBMS), Distributed Systems & Cloud Computing, Computer Networks & Internet Protocols, Theory of Computation & Automata, Compiler Design, Software Engineering & System Analysis, Computer Architecture, Artificial Intelligence, Microprocessors & Interfacing, Discrete Mathematics, and Network Security & Cryptography.